Immersive Chemistry Lab | User Testing IP2b

These are the four user-supplied transcript texts, retained as evidence rather than rewritten findings. P1-P4 may contain apparent speech-to-text errors

Participant 1 -

Researcher: Hi, how are you doing?

Participant: Hi, I'm good How are you?

Researcher: I'm good too. Thank you for taking part in this VR prototype experience. May I have your consent to record this audio?

Participant: Yeah, sure.

Researcher: Okay, thank you. So this is an immersive chemistry lab that I've worked on. This is a VR app, as you can see. And in this experience, you will basically explore a virtual chemistry lab and perform a few interactive activities or experiments as you can call them. And I'm testing how understandable and comfortable this whole experience is for you. So feel free to, you know, explore the whole room, the whole chemistry lab, and I'll record your experiences. Is that okay?

Participant: Yeah that's fine.

Researcher: Okay, uh, now you can start uh, your uh, like using the app and just uh, think out loud and I'll record the experience.

Participant: Yeah, sure. All right, so I see like some instructions, I'm just gonna quickly reap them. Alright, okay. Alright, so I see like 2 speakers here and a pond. Yeah, I'm too pour into the cold round, trace them. Okay, so I'm gonna do that.

Researcher: Okay, uh, now you can just rotate the the beaker in inside the pot.

Participant: Okay, okay, got it.

Researcher: Yeah, you can also read the instructions that are given here.

Participant: Yeah, yeah, I'm just trying to like. I'm rotating the bonds right now. Yeah. Okay. No, okay, so there's like a reaction happening. Right?

Researcher: Yeah. And now, Yeah, you can you can see like open the molecular boat to look at.

Participant: Okay.

Researcher: So here you can see that you can trace the.

Participant: Yeah, yeah, yeah. I'm just trying to trace it. Okay.

Participant: Oh, okay, okay, I traced it alright, so now I'm in the molecular world, right?

Researcher: Yeah. Okay, so see how you can, like, just explode it and see if you can interact with the objects.

Participant: Okay, okay. I think they do hydrogen atoms, okay, so I have to like connect these. Alright? So now it's like a water molecular molecule formed, right?

Researcher: Yeah, yeah.

Participant: Okay. So now, what can happen?

Researcher: Yeah, you can, like, separate it or play around with it, like, just see how you can..

Participant: So, like, I don't see a major reaction happening, but, uh,

Researcher: yeah, actually, this is like a initial prototype. So right now you only have like 3 molecules that you can play around with, but obviously I will like introduce more molecules and more complex chemical reactions later.

Participant: Oh, okay, okay, okay, got it.

Researcher: Yeah. Okay, now you can, like, uh, press B to exert this area.

Participant: Okay, so. All right, so I'm back in this area.

Researcher: Yeah.

Participant: Yeah, so I'm assuming this is like the end?

Researcher: Uh, no, you can just explore the lab and see like if there's any other experiments. Like I told you, like 3 experiments you can perform in this lab.

Participant: Mm-hmm. Okay. Yeah. So I'm just gonna move around. Okay, so experiment magical mixing. All right, so, grab the buttons. I'm gonna grab the wooden spoon.

Researcher: Uh, yeah.

Participant: Okay, so I'm gonna pick the spoon now and make, then place it inside the bowl.

Researcher: Uh, yeah, you can just, yeah, rotate it like this.

Participant: All right, so, okay.

Researcher: Ah, now you can see the colour changed, right?

Participant: Okay, okay, huh. Yes, yes, yes. Okay, I can see it.

Researcher: Yeah, so this was, like, the 2nd experiment, and now you can move around and see how you can...

Participant: I mean, is there any other experiment that is left? All right, sounds still moving on. Mm. Mm. Okay, so I see another experiment happening. Heating reaction. Okay, grab the blue flask, hold it above the burnage, and you rotate, because I'm gonna do what the instructions say.

Researcher: Yeah. Now, you can, as you can see, in the instruction, it says gently rotate. So now you can.

Participant: Okay, so I'm gonna do that.

Researcher: Yeah, so now you can see like the reaction happened, right?

Participant: Yes, yes, yes. There's like a reaction in here.

Researcher: Okay. So... Yeah. So this was like the end of the whole experience.

Participant: Okay.

Researcher: Yeah. Now I would like to ask just a few questions before we end this. Did you think that the whole experience felt natural and you were able to understand the instructions without my guidance or did you think that there was some guidance needed here and there?

Participant: I think the whole experience was, according to, like, a VR experience, it was, it felt natural. The instructions were okay as well. I think you can also add a little bit more, take a bit more information about the reactions that are taking place. in the molecular world or like maybe like a transition or something. Except for that, I think the whole experience was very natural.

Researcher: Okay. And did you think that there was something that felt unnatural or just felt very awkward, especially during the VR experience, you know, like wearing the headset, the world felt weird or you felt dizzy or something else?

Participant: No, no, I think it was very natural. I didn't find anything problematic. Uh, just the instructions, you can add a bit more information about the actual experiments. That's it.

Researcher: Okay. And did you think that this looks better while you were wearing the headset or did it look weird when you were in that chemistry lab? Like, were you able to understand that this is a chemistry lab?

Participant: Yeah, I could understand that this was like meant to be a mystery lab. I think it's a different aesthetic, but it's nice. It's not, uh, it has a nice like new attach to it.

Researcher: Okay, okay. Thank you so much. This is where the whole experience ends. Thank you so much for your time and yeah, this interview. you.

Participant: Okay, thank you. Thank you.

Participant 2 -

Researcher: Hi, how are you doing?

Participant: Hi, yeah, I'm good. How are you?

Researcher: I'm good too, thank you. And, uh, thanks for helping me test this VR prototype. Before we get started, can I just check, are you okay with me recording the audio?

Participant: Yeah, sure. That's fine.

Researcher: Okay, great. So this is an immersive chemistry lab that I've worked on. You'll basically get to look around and try three experiments. I'm mainly checking whether the instructions make sense and whether the whole thing feels comfortable to use. So, um, as you're doing it, just tell me what you're seeing or thinking. Even if you're not sure what to do, that's completely fine. Okay?

Participant: Yeah, yeah. Okay, I'll try to say what I'm doing.

Researcher: Alright, whenever you're ready, you can start.

Participant: Okay... so, hang on, I'm just looking at the instructions first. Um, right. Two beakers, and I need to pour them into this cauldron. Okay, yeah. I can see them. I'll just... I'll try picking this one up first.

Researcher: Yeah, go ahead.

Participant: Okay, I've got the first one. Uh, wait, I'm sort of trying to get it over the cauldron. Right. Now do I, like, turn my hand? Like this? Hmm, okay, give me a second.

Researcher: Yeah, just rotate the beaker towards the cauldron.

Participant: Oh, okay. Yeah, yeah, I see. Right, let me move it a little closer... okay. Now the second one. Wait... alright, I've got it. Just turning it now. Oh! Okay, yeah, something's happening.

Researcher: Yeah, so that's the reaction.

Participant: Right, right. Yeah, I can see that. Um, so do I do anything else with these, or is there another part?

Researcher: Yeah, if you look at the next instruction, there's a molecular level you can enter.

Participant: Oh, okay, wait, let me read that properly. 'Trace the circle around the portal.' Right. So I just sort of follow the circle with my hand? Okay, I think I get it.

Researcher: Yeah, you can try tracing around it.

Participant: Okay, so I'm gonna start here... and go around. Hmm. Wait. I don't think that did anything. Maybe I went a bit too quickly? Let me just try it again.

Researcher: Yeah, that's okay. Just follow the circle around the portal.

Participant: Okay, yeah. Just going around it now... oh, there we go! Okay. That worked. So, um, I'm in the molecular world now, yeah?

Researcher: Yeah, that's right. Just explore what's in front of you.

Participant: Okay, I'm just looking around. There are, um... three atoms here, I think? Two hydrogen and one oxygen. So, right, I'm guessing I need to connect them. Let me see if I can grab one.

Researcher: Yeah, go ahead and see if you can bring them together.

Participant: Alright, I've got this one. Wait, it moves with my hand, okay. I'm bringing it closer to the oxygen... little bit closer... oh, okay, it connected. Right. Now the other hydrogen. Hang on... okay, yeah. There we go. So that's a water molecule now, I think.

Researcher: Yeah, exactly.

Participant: Okay, nice. Wait, can I pull these apart again, or are they just connected now?

Researcher: Yeah, you can try pulling them apart and see what happens.

Participant: Okay, let me try... just pulling this one away. Oh, yeah, it separated. Okay, that's nice. I actually like that bit, because, um, you can move the atoms yourself instead of just looking at them.

Researcher: Okay. Whenever you're ready, press B to return to the lab.

Participant: Alright, B... okay, there we go, I'm back. Um, right, so you said there are two more experiments somewhere in here?

Researcher: Yeah, just look around the room.

Participant: Okay, I'll just look around. Um... oh, here. Magical mixing. Let me read this properly. Okay, pick up the wooden spoon and stir inside the bowl. Alright, that sounds simple enough. I'll try.

Researcher: Yeah.

Participant: Okay, so I've got the spoon. Uh, I'm putting it into the bowl now. And, um, just moving it around like this? Wait, I can't really tell if it's doing anything yet. Should I keep going?

Researcher: Yeah, just keep stirring for a little bit.

Participant: Okay, yeah, just going around again... oh, right, the colour's changed. Yeah, yeah, I can see it now. Okay, that one was quite easy once I got it in the bowl.

Researcher: Alright, and there's one more experiment.

Participant: Okay, let me see... I'm just moving around a bit. Oh, okay, heating reaction. Right. It says grab the blue flask, hold it above the burner and gently rotate. Okay, so I'm picking it up... wait, let me position it properly. Alright, I think that's right.

Researcher: Yeah, now try gently rotating it.

Participant: Okay... I'm turning it a little. Hmm... oh, okay, yeah, I can see something happening now. Right, so that's the reaction. Okay, got it.

Researcher: Yeah, that's the last one. So, um, I just have a few questions before we finish. Did the instructions make sense on their own, or were there places where you needed me to explain things?

Participant: Um, mostly they made sense, yeah. Like, the pouring and the mixing were quite straightforward once I started doing them. With the portal, I understood what it was asking, but, uh, when nothing happened the first time, I wasn't really sure if I'd traced it correctly. So I sort of had to guess and do it again.

Researcher: Okay, okay. And what about picking things up or moving the atoms? Did that feel natural to you?

Participant: Yeah, I think so. At first I was just trying to work out how to hold the beaker and rotate it, you know? But after a few seconds it was fine. The atoms were actually easier than I expected. I could just move them closer, and they connected.

Researcher: Okay. And did you feel dizzy or uncomfortable at any point while wearing the headset?

Participant: No, no, not really. I was okay. I didn't feel dizzy or anything like that.

Researcher: Okay. And what did you think of the lab itself? Could you tell it was supposed to be a chemistry lab?

Participant: Yeah, yeah, definitely. It's got this, um, kind of magical look rather than a normal school chemistry lab, but I think that works. I could still tell it was about doing experiments.

Researcher: Okay. And is there anything you think I should improve?

Participant: Um... probably make it a bit clearer when the portal circle has actually been traced properly. Because that was the part where I wasn't sure. And maybe, yeah, a little more explanation after each reaction. I could see something had happened, but I didn't always know what it meant in terms of chemistry.

Researcher: Okay, alright. That's really helpful. Thanks so much for your time and for trying it out.

Participant: Yeah, no worries. Thank you.

Participant 3 -

Researcher: Hi, how are you?

Participant: Hey, I'm good. How are you?

Researcher: Yeah, I'm good too. Thanks for coming to test this. Before we start, just quickly, is it okay if I record the audio of the session?

Participant: Yeah, that's okay.

Researcher: Okay, great. So this is a VR chemistry lab prototype with three experiments in it. I'm looking at whether the instructions are clear and how the interactions feel. You don't need to rush, and you don't need to get everything right. Just, uh, say what you're thinking as you go, even if it's something really small. Is that alright?

Participant: Yeah, sure. I might go a little slowly because I'm trying to read everything, but yeah.

Researcher: That's completely fine. You can start whenever you're ready.

Participant: Alright, so, um... okay, I'm in the lab. Just looking around. Oh, there's an instruction here. Let me read it... okay, two beakers into the cauldron. Yeah, I can see those. I'm gonna pick up this one first.

Researcher: Mm-hmm.

Participant: Okay, I've got it. So I'm just, uh, lifting it a little and then rotating it. Right... okay. And now the other one. Hang on, let me bring it closer so it's actually over the cauldron. Okay... turning it now. Oh, something changed. So that's the reaction, I guess?

Researcher: Yeah, that's the reaction.

Participant: Alright. Okay, let me see what the next bit says. Something about a portal... yeah. 'Trace the circle around the portal to enter the molecular level.' Okay. I can see the circle there. I think I just follow it round, right?

Researcher: Yeah, you can give that a try.

Participant: Okay, so I'll start here... just following it around... and, oh! Yeah, I'm in. Okay, that was quicker than I thought. That transition's pretty cool, actually.

Researcher: Okay, have a look around and see what you can interact with here.

Participant: Right. Um, okay, so there are three atoms here. I think, yeah, two hydrogen and one oxygen. And, uh, I'm guessing I connect them to make water. I'll just try grabbing one of these.

Researcher: Yeah, you can try that.

Participant: Alright, so I've got the hydrogen. It's moving, okay. I'm bringing it towards the oxygen... okay, there, it's joined. And now this one. Hang on, I just need to get it close enough... right, yeah. That's H2O now.

Researcher: Yeah, exactly.

Participant: Okay, nice. Um, let me see... can I do anything else here, or is it just joining them together?

Researcher: You can try separating the atoms as well.

Participant: Oh, okay. Let me pull this one away... wait, just a little further... yeah, it broke apart. Okay, I like that. It sort of shows how they're connected. I think that part makes sense visually.

Researcher: Alright. Press B whenever you want to return to the lab.

Participant: Okay, B... right, I'm back. So I'll just look for the next one. Um, let me turn around a little... oh, magical mixing. I can see a spoon and a bowl here.

Researcher: Yeah.

Participant: Okay, it says grab the wooden spoon and stir. So I'm picking it up now... just making sure I've got it. Okay. I'm putting it in the bowl and moving it around. Hmm... oh, there it goes, the colour changed. Right, that was pretty straightforward.

Researcher: Okay, and there's one more experiment in the room.

Participant: Alright, I'll have a look. Um... okay, heating reaction. Let me read. Grab the blue flask, hold it over the burner, gently rotate. Right. So I'm lifting it up... moving it over the burner... I think that's close enough. And turning it a little. Oh, yep, I can see the reaction now.

Researcher: Yeah, that's the last one.

Participant: Okay. So that's all three, yeah? I think I've done everything.

Researcher: Yeah, that's the whole experience. So, um, just a few questions. How did you find the instructions overall? Was anything unclear?

Participant: I think they were pretty easy to follow. I liked that the portal one actually told me to trace the circle, so I didn't have to guess what the gesture was. I did stop and read it properly, but, yeah, I didn't really get stuck anywhere.

Researcher: Okay. And using the controllers, picking up and moving the objects, did all of that feel natural?

Participant: Yeah, mostly. With the very first beaker I was just figuring out, like, how much I needed to turn it. But after that the actions made sense. Stirring the bowl was probably the simplest, because it's just what you'd expect to do.

Researcher: Okay. And did anything feel awkward or uncomfortable, especially with the headset?

Participant: No, not really. I didn't feel dizzy or anything. I was comfortable, yeah.

Researcher: Okay. And when you first entered, could you tell what kind of environment you were in?

Participant: Yeah, I could. It's obviously a chemistry lab, just more of a fantasy version. Um, I think that's actually a nice touch. It doesn't feel like a normal classroom, but it still makes sense as a place to do the experiments.

Researcher: Okay. And if you could improve one thing about it, what would you pick?

Participant: Probably, um, some more explanation about what the reactions mean. Like, with the heating experiment, I did what it said and saw the change, but I wasn't sure what chemistry concept it was showing. The molecular level was a bit clearer to me because I could see the hydrogen and oxygen connecting.

Researcher: Okay, yeah. Anything else you'd like to mention?

Participant: Um... no, I think that's about it. It was fun to try, yeah.

Researcher: Alright. Thanks so much for taking part.

Participant: Yeah, thank you.

Participant 4 -

Researcher: Hi, how are you doing today?

Participant: Hi, I'm good, thanks. How are you?

Researcher: I'm good too. Thanks for joining. So, um, before we begin, would you be comfortable with me recording the audio during the session?

Participant: Yeah, that's fine.

Researcher: Okay, thank you. So this is an immersive chemistry lab in VR. You can explore it and do three small experiments. I'm trying to see whether the instructions are understandable and whether the interactions feel natural. As you go through it, just tell me what you're doing, what you're looking at, or if you're not sure about anything. There's no need to hurry. Okay?

Participant: Yeah, okay. I'll try to talk through it.

Researcher: Alright, you can start whenever you're ready.

Participant: Okay, um... I'm just looking around first. It's got a sort of magical lab feeling. Right, there's an instruction here. Let me get a better look... okay. So I need to take the two beakers and pour them into the cauldron. Alright, let me try.

Researcher: Yeah, go ahead.

Participant: Okay, so I've picked up the first one. Wait, I'm trying to move it over the cauldron without, uh... okay, I think that's fine. Now turning it. Right, okay. And the second one... hang on, I've got it. Let me move it closer. Okay, just tilting it now. Oh, yeah, I can see something happening.

Researcher: Yeah, that's the reaction.

Participant: Okay. Um, so what's next? Oh, wait, there's another instruction here. Let me read it. 'Trace a circle around the portal to enter the molecular level.' Right... um, where's the portal? I'm just looking... oh, there it is. Okay, I can see the circle.

Researcher: Yeah, see if you can follow that circle.

Participant: Okay, let me try. I'm starting here... just going around it... hmm. Nothing happened. Wait, did I go the wrong way? Or maybe I didn't follow it closely enough. I'll just try again.

Researcher: Yeah, that's okay. Try following the circle around the portal again.

Participant: Okay, right. So, slowly this time... just around here... oh, there we go! Yeah, now I'm in. Okay, so this is a different area. Got it.

Researcher: Yeah, this is the molecular level. You can explore the atoms here.

Participant: Alright. Just looking at what's in front of me. So, um, there are three atoms. These are hydrogen and oxygen, right? I think I've got that.

Researcher: Yeah.

Participant: Okay, so I should connect the two hydrogen atoms to the oxygen. Let me grab this one... wait, I'm just trying to get hold of it. Okay, yeah, it's moving now. I'm bringing it closer... oh, it connected. Right. Now the second one. Let me move it over... there. Okay, so that's water now.

Researcher: Yeah, you've formed a water molecule.

Participant: Okay. Um, is that everything in here, or can I do something else with it?

Researcher: You can also try pulling the atoms apart.

Participant: Oh, okay, let me try. I'm grabbing one again... just pulling it away... oh, yeah, it separated. Right, that's interesting. So you can actually see the bond being made and then broken. I like that.

Researcher: Okay. You can press B when you're ready to return to the lab.

Participant: Alright, B... okay, I'm back. Right. There were two more experiments, weren't there? I'll have a look around.

Researcher: Yeah, you can look around for them.

Participant: Okay... um, magical mixing. Yeah, I can see that one. It says use the spoon. So I'm reaching for the wooden spoon now... okay, I've got it. And I put it inside the bowl and stir, right?

Researcher: Yeah.

Participant: Okay, I'm stirring now. Hmm. I can't really see anything yet. Wait, do I need to go faster or do a bigger circle? Let me keep trying... oh, okay, the colour changed. Yeah, I can see it now.

Researcher: Yeah, that's the reaction for that experiment.

Participant: Okay, nice. Right, so that one makes sense. I just needed to keep stirring a little longer.

Researcher: Yeah, there's one experiment left.

Participant: Okay, let me find it. Um... oh, heating reaction. Right, so I'm reading the instructions. Pick up the blue flask and hold it above the burner. Okay, I've got it. Let me move it over here... and it says gently rotate. Just turning my wrist a little... wait... oh, yeah, now I can see the reaction.

Researcher: Yeah, that's the final experiment.

Participant: Okay, right. I think I've done all of them then.

Researcher: Yeah, that's the end of the experience. Just a few questions before we finish. Were you mostly able to follow the instructions on your own?

Participant: Um, mostly, yeah. I think the mixing and heating instructions were quite clear, because they told me exactly what action to do. With the portal, I knew I had to trace the circle, but when it didn't work at first, I couldn't really tell if I'd done it wrong. I sort of just tried it again more slowly.

Researcher: Okay. And was there any interaction that felt unnatural or difficult?

Participant: I wouldn't say difficult, exactly. It was more that I needed a moment to figure some things out. Like, getting the first beaker into position, and then the portal circle. Once I got those, it felt okay. I knew what I was doing.

Researcher: Okay. Did you feel dizzy or uncomfortable at any point while wearing the headset?

Participant: No, no. I was fine. Didn't feel dizzy or uncomfortable.

Researcher: Okay. And what did you think of the lab visually? Did it make sense as a chemistry lab?

Participant: Yeah, I could tell it was a chemistry lab. It doesn't look like a regular school one, obviously, but, um, I like that it's a bit magical. It makes you want to look around the room, I think.

Researcher: Okay. And if you could change anything about the experience, what would you suggest?

Participant: I think, um, make it a little clearer when the portal tracing has worked. Because I wasn't sure the first time. And maybe a bit more information about the reactions, like what actually happened after you do the action. I could perform them, but I didn't always understand the chemistry behind what I was seeing.

Researcher: Okay, that's useful. Anything else you wanted to add?

Participant: Um... no, I think that's it really.

Researcher: Alright. Thank you so much for testing the prototype and giving your feedback.

Participant: Yeah, no worries. Thank you.